

Operations Research & Applications programming

Ch. 3

The Simplex Method and Sensitivity Analysis

Introduction

- *This chapter*, This chapter details the simplex method for solving the general LP problem. It also explains how simplex-based sensitivity analysis is used to provide important economic interpretations about the optimum solution, including the *dual prices* and the *reduced cost*.

Introduction

- ❑ The **Simplex Method** attempts to move from one corner point of the solution space to a better corner point *until the optimum is found*.
- ❑ When you complete this chapter, you will be in a position to read and interpret the output reports provided by commercial software and our TORA software.

Algebraic Solution of LPPs – Simplex Method

- ❑ To solve an LPP algebraically, we first put it in the **Standard Form**. This means
 1. All decision variables are **nonnegative**.
 2. All constraints (with the exception of the non negativity of the variables) are **equations with nonnegative RHS**.
- ❑ These two requirements are imposed here primarily streamline calculations.

Converting inequalities into equations

- *Example 3.1* Consider the LPP

$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to

$$x_1 + 3x_2 \leq 6$$
$$3x_1 + 2x_2 \leq 6$$
$$x_1, x_2 \geq 0$$

- We make the $\leq$ inequalities into equations by adding to each inequality a “slack” variable (which is nonnegative). Thus the given LPP can be written in the equivalent form.

Converting inequalities into equations

□
$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to :
$$x_1 + 3x_2 + s_1 = 6$$

$$3x_1 + 2x_2 + s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

s_1, s_2 are *slack* variables

□ ***Thus***, we seem to have complicated the problem by introducing two more variables; but then we shall see that this is easier to solve. This is one of the “beauties” in mathematical problem solving.

Converting inequalities into equations

- The $\geq$ inequalities are made into equations by subtracting from each such inequality a “surplus” (non-negative) variable.
- *Thus*, the LPP

$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to

$$x_1 + 3x_2 \leq 6$$
$$3x_1 + 2x_2 \geq 6$$
$$x_1, x_2 \geq 0$$

Converting inequalities into equations

is equivalent to the LPP

Subject to *Maximize* $z = 2x_1 + 3x_2$

$$x_1 + 3x_2 + s_1 = 6$$

$$3x_1 + 2x_2 - s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

s_1 is a *slack* variable; and s_2 is a *surplus* variable.

Converting inequalities into equations

- If in a constraint, the RHS *constant is negative*, we make it positive by multiplying the constraint by -1.
- *Thus*, the LPP

$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to

$$x_1 + 3x_2 \leq 6$$

$$3x_1 + 2x_2 \geq -2$$

$$x_1, x_2 \geq 0$$

Converting inequalities into equations

- Is equivalent to the LPP

$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to

$$x_1 + 3x_2 \leq 6$$

$$-3x_1 - 2x_2 \leq +2$$

$$x_1, x_2 \geq 0$$

Converting inequalities into equations

□ It's standard form is the LPP

$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to

$$x_1 + 3x_2 + s_1 = 6$$

$$-3x_1 - 2x_2 + s_2 = 2$$

$$x_1, x_2, s_1, s_2 \geq 0$$

s_1 and s_2 are a *slack* variable.

Dealing with unrestricted variables

- If, in a LPP, a decision variable x_i is unrestricted (in sign) i.e. it can take positive as well as negative values, then we can “by writing $x_i = x_i^+ - x_i^-$ where x_i^+, x_i^- are (*defined below*) nonnegative” make the LPP into an equivalent LPP where all the decision variables are ≥ 0 .

- *Not* that
$$x_i^+ = \frac{|x_i| + x_i}{2}; \quad x_i^- = \frac{|x_i| - x_i}{2}$$
$$x_i = x_i^+ \text{ if } x_i \geq 0 \text{ and } -x_i^- \text{ otherwise}$$

Dealing with unrestricted variables

□ *Thus*, the LPP

$$\text{Maximize } z = x_1 + 3x_2$$

Subject to

$$x_1 + x_2 \leq 2$$

$$-x_1 + x_2 \leq 4$$

$$x_1 \text{ unrestricted}, x_2 \geq 0.$$

Converting inequalities into equations

□ Is equivalent to the LPP

$$\text{Maximize } z = x_1^+ - x_1^- + 3x_2$$

Subject to

$$x_1^+ - x_1^- + x_2 + s_1 = 2$$

$$-x_1^+ + x_1^- + x_2 + s_2 = 4$$

$$x_1^+, x_1^-, x_2, s_1, s_2 \geq 0$$

s_1 and s_2 are a *slack* variable.

Translation from Graphical to Algebraic Solution

- ❑ Consider a LPP (in standard form) with m constraints and n decision variables. We assume $m \leq n$.
- ❑ We choose $n - m$ variables and set them equal to zero. Thus we will be left with a system of m equations in m variables.
- ❑ If this $m \times m$ square system has a unique solution, this solution is called a *basic solution*. Further if it is feasible, it is called a *Basic Feasible Solution* (BFS).

Translation from Graphical to Algebraic Solution

- ❑ The $n-m$ variables set to zero are called *nonbasic* and the m variables which we are solving for are known as *basic variables*.
- ❑ Thus a basic solution is of the form $\mathbf{x} = (x_1, x_2, \dots, x_n)$ where $n-m$ “components” are zero and the remaining m components form the unique solution of the square system (formed by the m constraint equations).
- ❑ **Note** that we may have a *Maximum* number of (maximum number of corner points) basic solution.

$$C_m^n = \frac{n!}{m!(n-m)!}$$

Translation from Graphical to Algebraic Solution

□ *Example 3.2* Consider the LPP

$$\text{Maximize } z = 2x_1 + 3x_2$$

□ Subject to

$$x_1 + 3x_2 \leq 6$$

$$3x_1 + 2x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

Translation from Graphical to Algebraic Solution

□ This is equivalent to the LPP (in standard form)

$$\text{Maximize } z = 2x_1 + 3x_2$$

Subject to

$$x_1 + 3x_2 + s_1 = 6$$

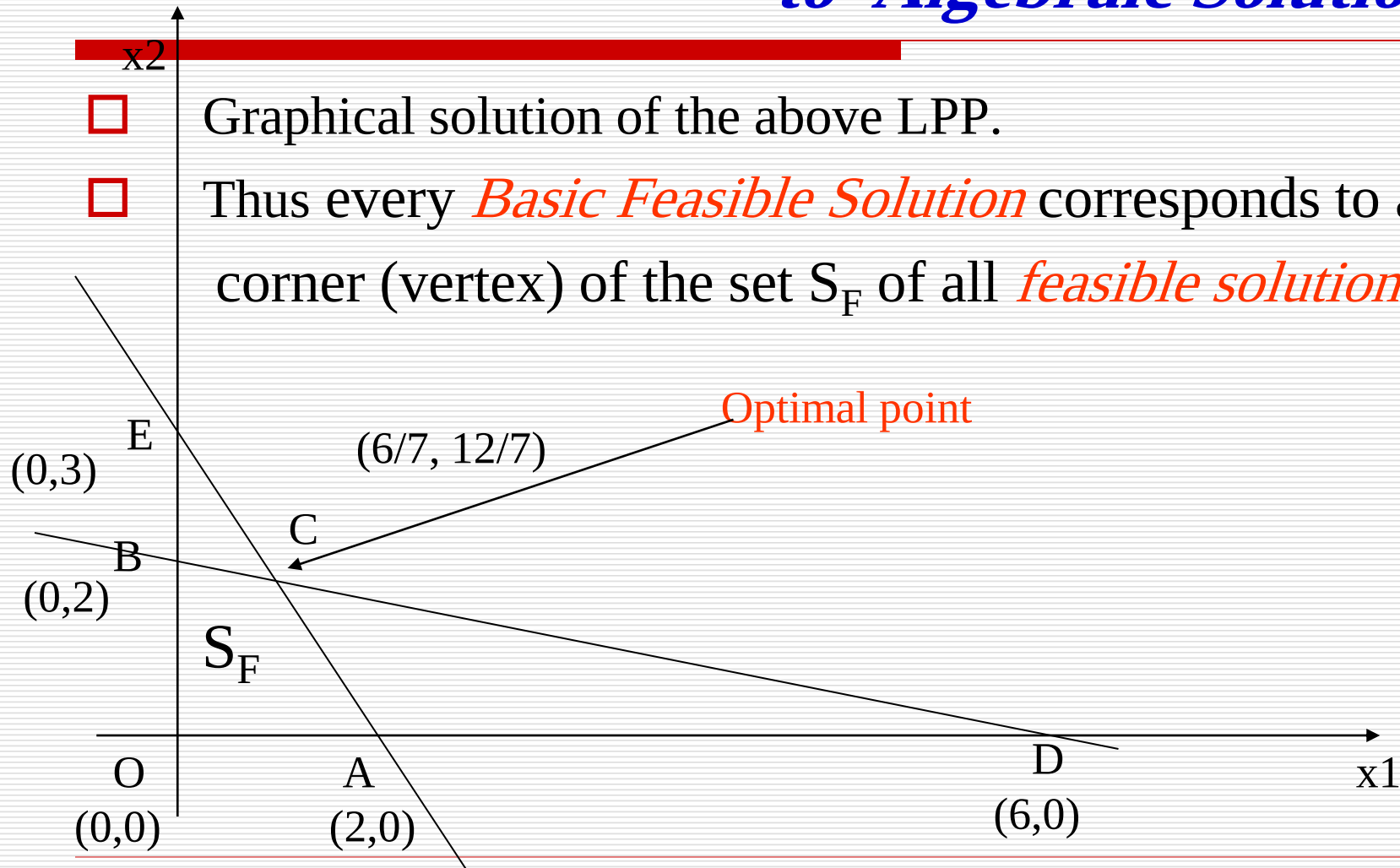
$$3x_1 + 2x_2 + s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

s_1, s_2 is a *slack* variable.

Translation from Graphical to Algebraic Solution

- Graphical solution of the above LPP.
- Thus every *Basic Feasible Solution* corresponds to a corner (vertex) of the set S_F of all *feasible solutions*.



Nonbasic (zero) variables	Basic variables	Basic solution (x_1, x_2, s_1, s_2)	Associated corner point	Feasible?	Objective value, z
(x_1, x_2)	(s_1, s_2)	(0,0,6,6)	O	Yes	0
(x_1, s_1)	(x_2, s_2)	(0,2,0,2)	B	Yes	6
(x_1, s_2)	(x_2, s_1)	(0,3,-3,0)	E	No	-
(x_2, s_1)	(x_1, s_2)	(6,0,0,-12)	D	No	-
(x_2, s_2)	(x_1, s_1)	(2,0,4,0)	A	Yes	4
(s_1, s_2)	(x_1, x_2)	$(\frac{6}{7}, \frac{12}{7}, 0, 0)$	C	Yes	48/7 Optimal

Example 3.3

- ❑ Question 5 (Problem set 3.2A – Page 90)

Consider the LPP

$$\text{Maximize } z = x_1 + 3x_2$$

Subject to

$$x_1 + x_2 \leq 2$$

$$-x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

$$x_1 \text{ unrestricted}, x_2 \geq 0.$$

Solution

□ The LPP Above is equivalent to the LPP

$$\text{Maximize } z = x_1^+ - x_1^- + 3x_2$$

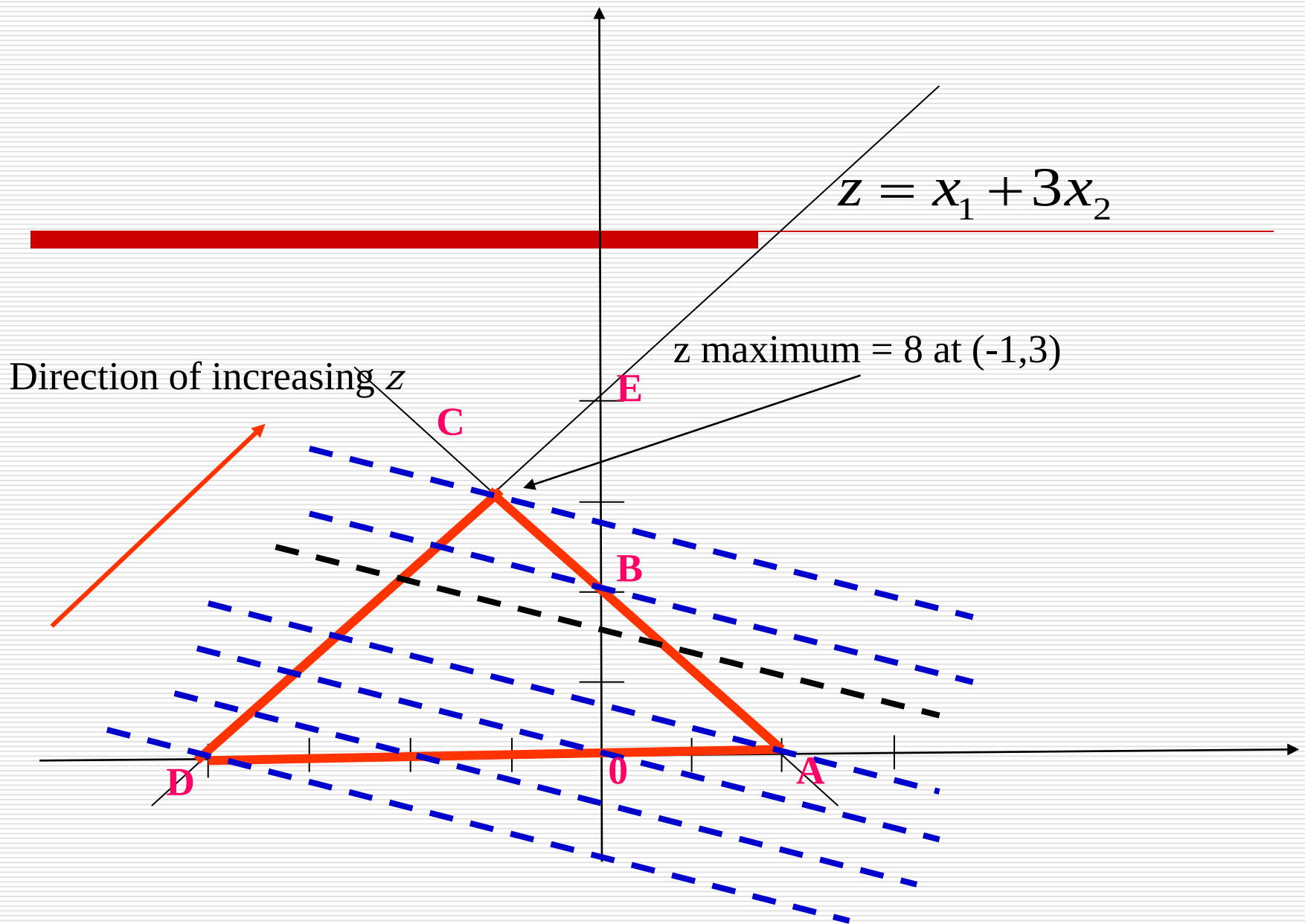
Subject to

$$x_1^+ - x_1^- + x_2 + s_1 = 2$$

$$-x_1^+ + x_1^- + x_2 + s_2 = 4$$

$$x_1^+, x_1^-, x_2, s_1, s_2 \geq 0$$

s_1, s_2 is a *slack* variable.



Nonbasic (zero) variables	Basic variables	Basic solution ($x_1^+, x_1^-, x_2, s_1, s_2$)	Assoc- iated corner point	Feasible?	Objective value, z
(x_1^+, x_1^-, x_2)	(s_1, s_2)	(0,0,0,2,4)	O	Yes	0
(x_1^+, x_1^-, s_1)	(x_2, s_2)	(0,0,2,0,2)	B	Yes	6
(x_1^+, x_1^-, s_2)	(x_2, s_1)	(0,0,4,-2,0)	E	No	-
(x_1^+, x_2, s_1)	(x_1^-, s_2)	(0,-2,0,0,6)	A	Yes	2
(x_1^+, x_2, s_2)	(x_1^-, s_1)	(0,4,0,6,0)	D	Yes	-4
(x_1^+, s_1, s_2)	(x_1^-, x_2)	(0,1,3,0,0)	C	Yes	8 <u>Optimal value</u>

Nonbasic (zero) variables	Basic variables	Basic solution $(x_1^+, x_1^-, x_2, s_1, s_2)$	Associated corner point	Feasible ?	Objective value, z
(x_1^-, x_2, s_1)	(x_1^+, s_2)	(2,0,0,0,6)	A	Yes	2
(x_1^-, x_2, s_2)	(x_1^+, s_1)	(-4,0,0,6,0)	-	No	-
(x_1^-, s_1, s_2)	(x_1^+, x_2)	(-1,0,3,0,0)	-	No	-
(x_2, s_1, s_2)	(x_1^+, x_1^-)	NO Solution	-	-	-

Hence note that the number of Basic Solutions can be less than $\binom{n}{m}$

Example 3.5

- Convert the following optimization problem into a LPP:

$$\text{Maximum } z = \max\{|2x_1 + 3x_2|, |3x_1 - 7x_2|\}$$

Subject to :

$$x_1 + x_2 \leq 2$$

$$-x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Example 3.5

□ Note that here the objective function is **NOT** linear. Let us put :

$$y = \max\{|2x_1 + 3x_2|, |3x_1 - 7x_2|\}$$

Hence

$$y \geq |2x_1 + 3x_2| \text{ and } y \geq |3x_1 - 7x_2|$$

Which it equivalent to :

$$y \geq 2x_1 + 3x_2, y \geq -(2x_1 + 3x_2)$$

$$y \geq 3x_1 - 7x_2, y \geq -(3x_1 - 7x_2)$$

Example 3.5

□ Hence the given optimization problem is equivalent to the LPP:

Maximum $Z = y$

Subjective to $x_1 + x_2 \leq 2$

$$-x_1 + x_2 \leq 4$$

$$2x_1 + 3x_2 - y \leq 0, -2x_1 - 3x_2 - y \leq 0$$

$$3x_1 - 7x_2 - y \leq 0, -3x_1 + 7x_2 - y \leq 0.$$

$$x_1, x_2, y \geq 0$$